MILITARY SPECIFICATION SHEET

RELAYS, SOLID STATE, SENSORS, ESTABLISHED RELIABILITY,
TYPE II (DC UNDERVOLTAGE SENSORS), CLASS C,
SPST (NO), HERMETICALLY SEALED (POTTED), PLUG-IN TERMINALS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-28894.

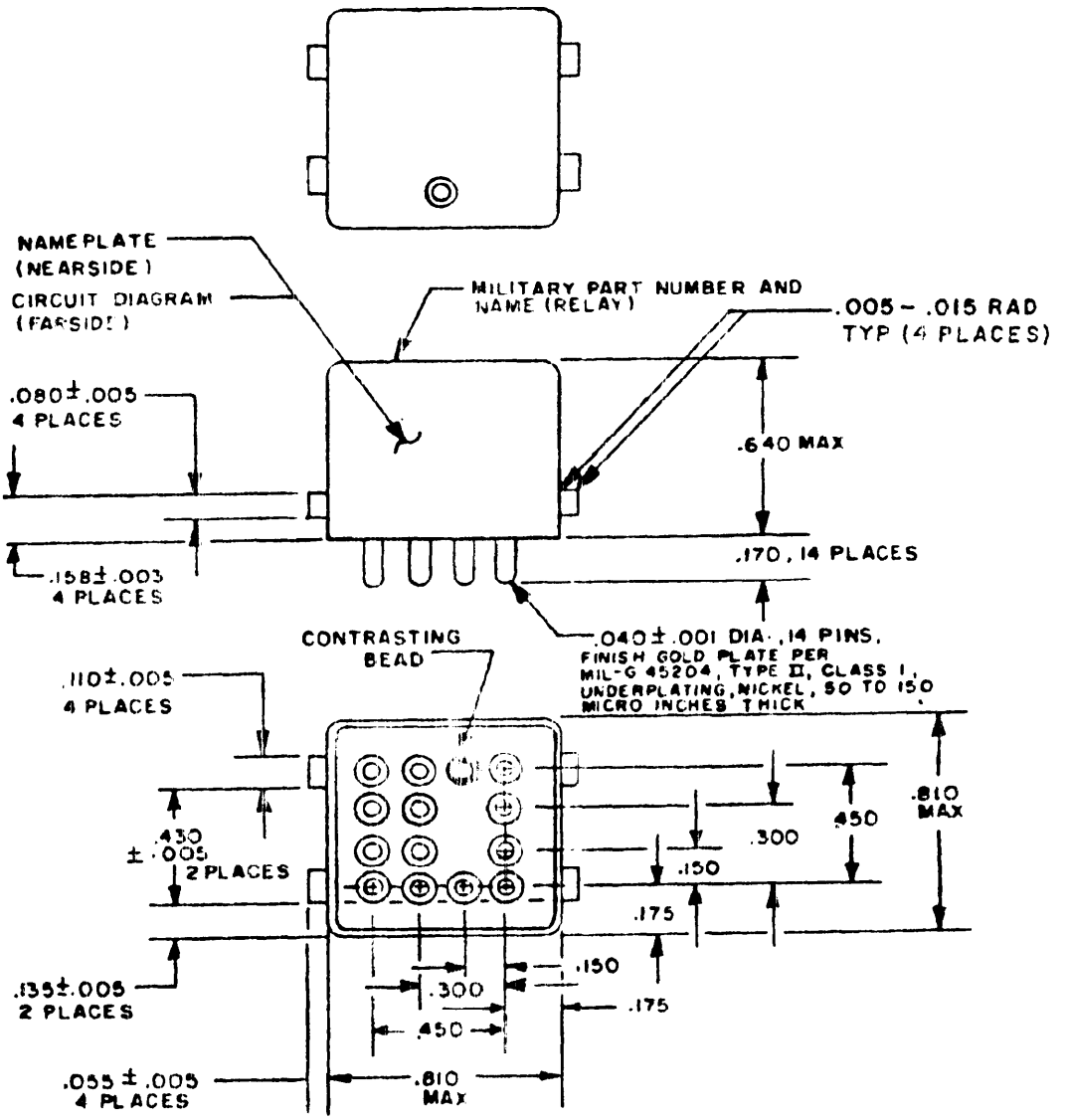
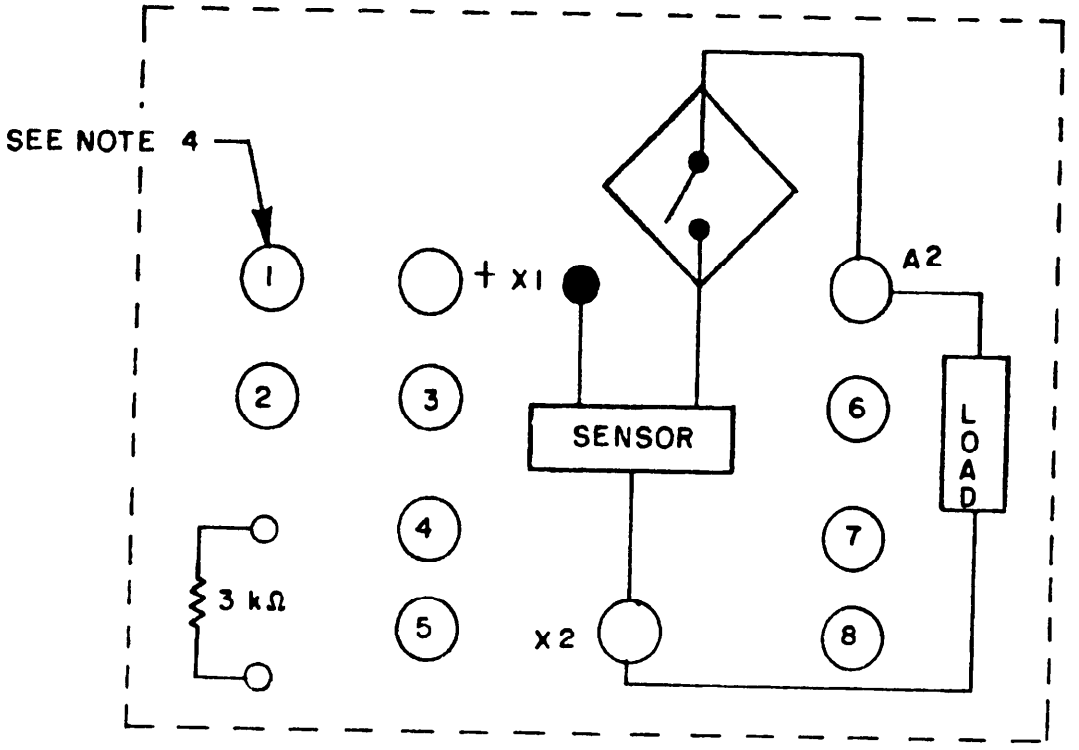


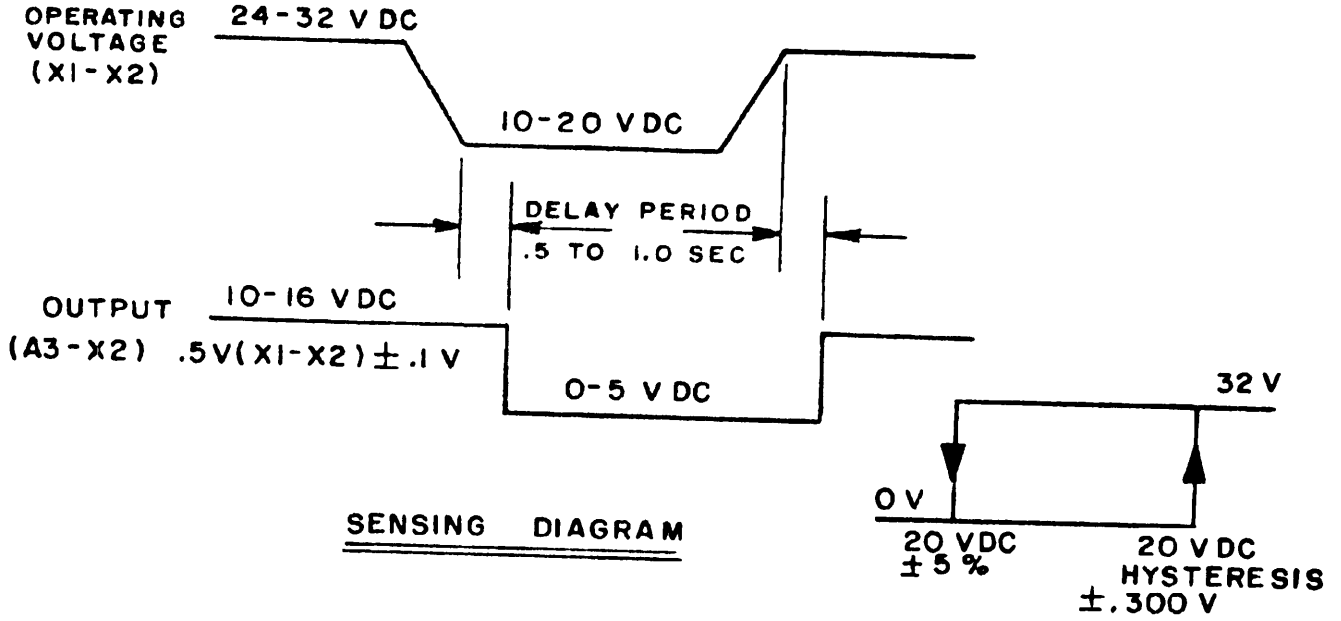
FIGURE 1. Outline dimensions and configuration.

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CIRCUIT DIAGRAM TERMINAL VIEW



SENSING DIAGRAM

LATCH HYSTERESIS

FIGURE 1. Outline dimensions and configuration - Continued.

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INCHES	MM	INCHES	MM
.001	0.03	.158	4.01
.003	0.08	.170	4.32
.005	0.13	.175	4.45
.015	0.38	.300	7.62
.040	1.02	.450	11.43
.055	1.40	.50	12.7
.080	2.03	.640	16.26
.110	2.79	.810	20.57
.135	3.43		
.150	3.81		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Terminal numbers shown above for reference only. Numbers do not appear on header.
4. Terminals 1, 2, 3, 4, 5, 6, 7 and 8 are non-functional leads.
5. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.

FIGURE 1. Outline dimensions and configuration - Continued.

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REQUIREMENTS:

Input data:

Load current: 16 mA, maximum.

Load impedance: 1 megohm, minimum.

Operating data:

Timing action: Delay on operate.

Time delay: Fixed; select from 0.5 second to 1.0 second.

Operating voltage: 10 to 32 V dc.

Sense voltage: Below 20 V dc ±5% to 10 V dc (over temperature range).

Output data:

Configuration: SPST, NO, Solid state closure to positive.

Output ratings (initial condition, above 20 V dc): One-half operating voltage ±.1 V (over temperature range).

Electrical data:

Transient specifications: 1/

Positive transients: 80 V in accordance with MIL-STD-704, figure G, limit 1.

Spike transients: ±600 V, 10 microseconds, maximum.

Self-generated transients: 50 V, maximum.

Spike susceptibility: +80 V maximum, -600 V minimum.

Electromagnetic interference: 2/ Per MIL-STD-461, class 1D.

Insulation resistance: 3/ 1,000 megohms minimum @ 500 V dc.

Dielectric resistance: 3/ 1,000 V rms, 60 Hz at sea level, 350 V rms @ 80 feet.

Timing cycle: Applicable.

Environmental data:

Ambient temperature:

Operating: -55°C to +85°C.

Storage: -55°C to +125°C.

1/ Transient specifications are based on a maximum duty cycle of 1/50.

2/ EMI test limits will not be exceeded during the timing interval, or when continuously energized under steady-state condition of MIL-R-28894.

3/ Dielectric withstanding voltage and insulation resistance measured between all mutually insulated terminals and between all terminals and case.

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Vibration:

Sinusoidal: 30G's, 10 to 3,000 Hz.

Random: 0.4 G^2 Hz power spectral density, 50 to 2,000 Hz in accordance with MIL-STD-810.

Shock: 50G's 11 ±1 ms, 1/2 sine, 3 axes.

Acceleration: 20G's in any axis.

Seal: Applicable.

Physical data:

Seal: Hermetic, potted.

Dimensions and configuration: See figure 1.

Terminations: See figure 1.

Terminal strength: 3 pounds pull.

Weight: 85 grams (3 ounces) maximum.

Finish:

Body: Bright tin plate.

Identification marking (full): Applicable.

Qualification inspection: See table I.

TABLE I. Qualification inspection and sample size. 1/

Single submission	Group submission	
18 units plus 1 unsealed unit for level L at C = 0	M28894/18-5000 M28894/18-1001	18 units plus 1 unsealed unit for level L at C = 0
34 units plus 1 unsealed unit for level M at C = 0		34 units plus 1 unsealed unit for level M at C = 0
Qualification inspection as applicable.		Qualification inspection as applicable.
	M28894/18-6000 M28894/18-7001 M28894/18-8000 M28894/18-9001	2 units each part number Qualification inspection, group 1 and shock.

1/ The number of units required for qualification testing will be increased as required in group V of the qualification inspection table of MIL-R-28894, if the supplier elects to test the number of units permitting one or more failures.

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Part number: Consists of the basic number of this specification sheet and a dash number denoting the required time delay, expressed in milliseconds. The first three digits are significant figures. The last digit is the number of zeros following the first three. A final letter designates the applicable failure rate level.

Failure rate level (percent per 1,000 hr.): L, 3.0; M, 1.0; P, 0.1; R, 0.01.

NOTE: Additional time delay relays within the 500 millisecond to 1.0 second delay range are available. Part number may be established as follows:

Examples: M28894/18-5000L equals a 500-millisecond time delay with "L" failure rate.

M28894/18-1003M equals a 1-second time delay with "M" failure rate.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AV
Navy - OS, SH

Preparing activity:
Navy - EC

Agent:
DLA - ES

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